

Association of Cl with C₂H₂ by unified variable-reaction-coordinate and reaction-path variational transition-state theory

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Contributed by Donald G. Truhlar, January 15, 2020 (sent for review November 14, 2019; reviewed by Stephen J. Klippenstein and Paul Marshall)

Barrierless unimolecular association reactions are prominent in atmospheric and combustion mechanisms but are challenging for both experiment and kinetics theory. A key datum for understanding the pressure dependence of association and dissociation reactions is the high-pressure limit, but this is often available experimentally only by extrapolation. Here we calculate the high-pressure limit for the addition of a chlorine atom to acetylene molecule (Cl + C₂H₂ → C₂H₂Cl). This reaction has outer and inner transition states in series; the outer transition state is barrierless, and it is necessary to use different theoretical frameworks to treat the two kinds of transition state. Here we study the reaction in the high-pressure limit using multifaceted variable-reaction-coordinate variational transition-state theory (VRC-VTST) at the outer transition state and reaction-path variational transition state theory (RP-VTST) at the inner turning point; then we combine the results with the canonical unified statistical (CUS) theory. The calculations are based on a density functional validated against the W3X-L method, which is based on coupled cluster theory with single, double, and triple excitations [CCSDT(Q)], and the computed rate constants are in good agreement with some of the experimental results. The chlorovinyl (C₂H₂Cl) adduct has two isomers that are equilibrium structures of a double-well C≡C–H bending potential. Two procedures are used to calculate the vibrational partition function of chlorovinyl; one treats the two isomers separately and the other solves the anharmonic energy levels of the double well. We use these results to calculate the standard-state free energy and equilibrium constant of the reaction.

kinetics | transition state | rate constant | electronic structure | direct dynamics

Radical–molecule reactions are central to many branches of chemistry, and such reactions dominate gas-phase combustion and atmospheric chemistry (1–4). Barrierless reactions are a special case of particular interest and provide great challenges for reaction kinetics theory (5–9). In the absence of a potential energy barrier, one must find the transition state by a free-energy criterion (10, 11), i.e., by variational transition-state theory (VTST) (12–14). The most widely applied form of VTST (12, 13), sometimes called reaction-path VTST (RP-VTST), is based on passage of reactive flux through a tight TS, by which we mean that the flux through the variationally best TS is in a region differing from a minimum energy path only by small-amplitude vibrations and possibly one or more one-dimensional internal rotations. However, this is not generally adequate at loose TSs such as encountered in barrierless association reactions (15–17). The state of the art for treating the loose TSs is variable-reaction-coordinate VTST (VRC-VTST) with a multifaceted dividing surface, as developed by Georgievskii and Klippenstein (18–20). Another complication is that such barrierless associations often have a tight inner TS as well as a loose outer one; the two TSs may need to be treated by the two different forms of VTST. The rate constant should be calculated taking account of both TSs (11, 21–27).

In the present article we study the addition reaction



where we carry out calculations for the condition where the pressure is high enough that the kinetics is in the pressure-independent second-order regime. The temperature and pressure dependence of reaction 1 have been studied both experimentally (28–36) and theoretically (36, 37). Experimental measurements of association rate constants generally do not go to high enough pressure to measure the high-pressure equilibrium limit directly, and this limit is obtained by theory-guided extrapolation. The kind of difficulty one faces in analyzing experimental data on association and dissociation reactions is well illustrated by the comments of Knyazev et al. (38) on the reaction system Cl + C₂H₄ ⇌ C₂H₄Cl, which is similar to reaction 1 studied here. Discussing their extraction of high-pressure-limit rate constants from the raw experimental data, they said, “One should bear in mind that the above model of the reaction is based on significant simplifications. . . . Different temperature dependencies of the collisional parameters and a different (within uncertainties) heat of formation of CH₂CH₂Cl will result in different values of rate constants. Moreover, the simplifications of the model (e.g., the assumption of a fixed TS and the representation of the tumbling motion of the C₂H₄ fragment relative to the Cl atom in the TS with low-frequency vibrational modes) result in uncertainties which are not easily estimated.” The present treatment of Cl + C₂H₂ → C₂H₂Cl attempts to overcome these sources of uncertainty by directly calculating the

Significance

A key issue in chemical kinetics is advancing the theoretical framework to handle reactions beyond the domain of textbook transition-state theory by including—for example—anharmonicity, barrierless transition states, transition states in series, and the effect of conformational flexibility on equilibrium constants. Other key issues are validating affordable electronic structure methods for direct dynamics and the direct calculation of high-pressure limiting rate constants, which are often obtainable experimentally only by extrapolation. The present article addresses all these issues for a prototype radical–molecule association reaction and thereby demonstrates how to combine improved theoretical methods to provide rate constants in cases where experimental data are uncertain or missing.

Author contributions: L.Z., D.G.T., and S.S. designed research; L.Z. and D.G.T. performed research; L.Z. and D.G.T. analyzed data; and L.Z. and D.G.T. wrote the paper.

Reviewers: S.J.K., Argonne National Laboratory; and P.M., University of North Texas.

The authors declare no competing interest.

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This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.1920018117/-DCSupplemental>.

First published March 2, 2020.

high-pressure limit rather than extrapolating to it, by using high-level electronic structure data to pin down the energetics, and by using VRC-VTST to eliminate the oversimplified treatment of the low-frequency vibrational modes.

We use the canonical unified statistical (CUS) (14, 27) theory to combine a VRC-VTST treatment of the outer TS with an RP-VTST treatment of the inner one to study the reaction in high-pressure limit. The microcanonical and E, J -resolved unified statistical models are also feasible, and might be more appropriate at pressures where the intermediate complex does not suffer enough collisions with the bath gas for thermalization (39–41).

The product, β -chlorovinyl, has the radical site on the α -carbon and the chlorine on the β -carbon (Fig. 1). This reaction is barrierless (36, 42) and serves as a prototype for barrierless addition of a doublet radical to a π -bond of an alkene or alkyne. Some previous closely related applications of VTST are studies of the addition reaction of OH to ethylene (39), the abstraction reaction of CN with ethane (40), the reactions of oxygen atoms with alkenes (41), and the addition of singlet methylene to acetylene (43). The only previous example we know of where VRC-VTST, RP-VTST, and CUS theory were applied together is the work of Monge-Palacios et al. (44) on $\text{HO}_2 + \text{propen-2-ol}$.

Methods and Theoretical Considerations. Electronic structure energies including nuclear repulsion but with neither zero-point energy nor vibrational-rotational thermal energies are potential energies for nuclear motion. The (classical) barrier heights and reaction energies are computed from potential energies. Adding zero-point energy gives the enthalpies of activation and enthalpies of reaction at 0 K. Finite-temperature enthalpies of activation, enthalpies of reaction, free energies of activation, and free energies of reaction are obtained by adding thermal contributions.

Validation of a Density Functional. The present calculations are carried out by direct dynamics (45, 46). In direct dynamics, one calculates the energy, gradient, and Hessian directly each time they are needed by the dynamics algorithm (rather than fitting the potential energy surface). Because direct dynamics is too expensive for the highest possible levels of electronic structure theory, our first step is to compare affordable Kohn–Sham density-functional theory to benchmark calculations at stationary points (saddle points and equilibrium structures) to select and validate a density-functional method (DFM) that is accurate for this particular reaction and that can be used for direct dynamics. A DFM is a combination of an approximate exchange-correlational functional and basis set. We tested two density functionals, M06-2X (47) and M08-HX (48), that were previously shown to perform quite well for main-group thermochemistry and kinetics. They

were combined with three polarized triple- ζ basis sets, jun-cc-pVTZ (49, 50), may-cc-pVTZ (49, 50), and MG3S (51).

The W3X-L method (52), which is a post-CCSD(T) method, where CCSD(T) denotes coupled cluster theory with single and double excitations and a quasiperturbative treatment of connected triple excitations was used to obtain a benchmark energy for each optimized geometry. The W3X-L protocol employs carefully selected basis sets to accurately approximate CCSDT(Q)/CBS results (coupled-cluster theory with single, double, and triple excitations and a quasiperturbative treatment of connected quadruple excitations with a complete basis set).

For each DFM, we optimized the geometries of all 14 stationary points (see *SI Appendix* for figures and tables specifying these points), and we ran single-point W3X-L calculations at those geometries. Comparing with benchmark results (detailed results for this step are in *SI Appendix*), we found that M06-2X/may-cc-pVTZ has the lowest mean unsigned error (MUE), and therefore we selected this DFM for direct dynamics calculations. The MUE is 0.66 kcal/mol, which is within the widely used criterion of 1 kcal/mol for “chemical accuracy.”

Validation of WMS. We also compared the W3X-L results to the more recent Wuhan–Minnesota scaling (WMS) (53) composite model chemistry method, and this comparison shows that the two methods agree quite well, with WMS being much less expensive.

Kinetics. All rate constant calculations correspond to equilibrated reactants and intermediates, and all chlorovinyl radicals produced are assumed to be stabilized. This assumption is valid at high enough pressures. The lower-pressure situation is not treated in the present study.

Reaction-path VTST. For the inner TS, we calculated the isoinertial-coordinate minimum energy path (54) (MEP) through the saddle point. This MEP is treated as the reaction path, and we performed vibrational analysis transverse to this path by using curvilinear coordinates (55); *SI Appendix*, Fig. S1. Then we searched along the reaction path to obtain the variationally best dividing surface, which is the variational TS (VTS). This produces the canonical RP-VTST rate constant, which can be written as

$$k_{\text{inner}} = \kappa \frac{k_B T}{h} \frac{\Phi^{\text{VTS}}}{\Phi^{\text{R1}} \Phi^{\text{R2}}} \exp\left(-\frac{V_{\text{MEP}}^{\text{VTS}}}{k_B T}\right), \quad [2]$$

where κ is the tunneling transmission coefficient, k_B is the Boltzmann constant, h is Planck’s constant, Φ is a partition function per unit volume (with the zero of energy at the local potential energy minimum) for the VTS or one of the reactants (R1 and R2), and $V_{\text{MEP}}^{\text{VTS}}$ is the potential energy of the VTS relative to reactants.

Eq. 2 can be rewritten in terms of the standard-state free energy of activation at the VTS $\Delta G_T^{\text{VTS,inner,o}}$ as

$$k_{\text{inner}} = \kappa \frac{k_B T}{h} \left(\frac{k_B T}{P^\circ}\right) \exp\left(-\frac{\Delta G_T^{\text{VTS,inner,o}}}{k_B T}\right), \quad [3]$$

where P° is the standard pressure (1 bar).

We replaced the DFM potential energies with the W3X-L benchmark ones in the calculation of k_{inner} to get more accurate results; we corrected the barrier for all variational dividing surfaces by using the correction obtained at the saddle point.

VRC-VTST. For loose transition states, it is difficult to define a useful reaction path because some of the internal coordinates have multidimensional wide-amplitude motion. We therefore use VRC-VTST with multifaceted dividing surfaces, as proposed by Georgievskii and Klippenstein (18–20), and in particular, we use the implementation (56) in the *Polyrate* (57) computer program. The reaction coordinate for the VRC scheme is defined by pivot

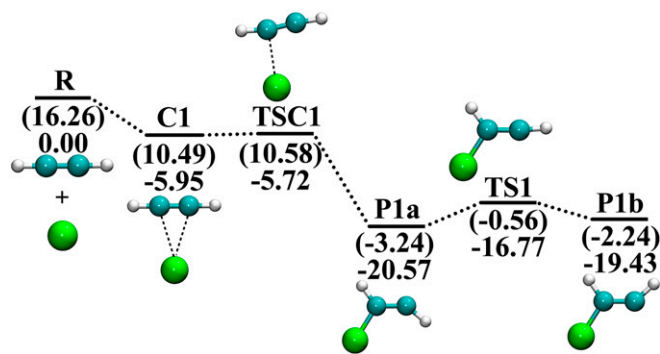


Fig. 1. M06-2X/may-cc-pVTZ stationary points (in kcal/mol) on the PES along the main association path for the $\text{Cl} + \text{C}_2\text{H}_2$ reaction with energies relative to the sum of the energies of the separated reactants. The values in parentheses include zero-point energies, i.e., they are relative enthalpies at 0 K.

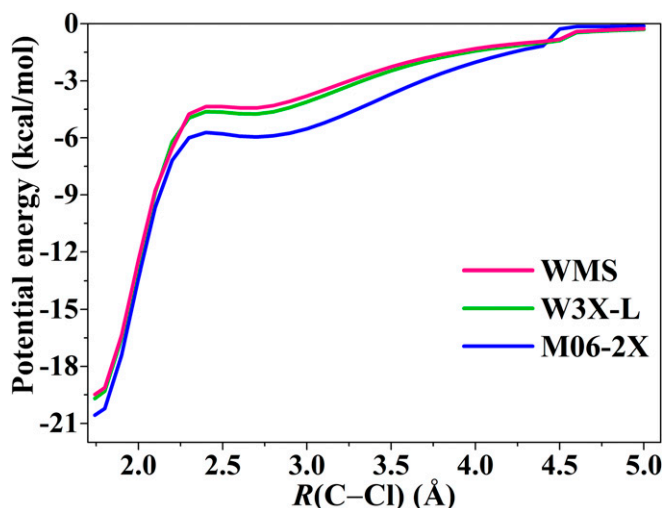


Fig. 2. Relaxed scan of electronic potential energies (in kcal/mol) along the C(1)–Cl distance (Å) calculated by M06-2X/may-cc-pVTZ and two high-level composite model chemistry methods (W3X-L and WMS); the zeros of the energies are set as the sum of the energies of separated reactants.

points as explained previously (14, 20, 58). We place one pivot point ($j = 1$) on the Cl atom and one on each C atom ($j = 2, 3$); see *SI Appendix*, Fig. S2. The reaction coordinate s is the minimal value of r_{12} and r_{13} , where r_{ij} is the distance between pivot points 1 and j . In this way the dividing surface has two facets. It would be interesting to study the dependence of the results on optimizing the pivot points or using a fixed center-mass separation instead of putting the pivot points on the reactive atoms.

For reactions with loose TSs, VRC-VTST divides the motions into conserved modes and transitional modes. The conserved modes are the vibrational modes of the separated fragments (reactants), and we assume the contribution of the conserved modes to the TS partition function cancels with the contribution of these modes to the reactant partition function. The transitional modes of the TS include the fragment rotations and the radial and orbital components of the relative translation; these are converted to vibrational modes and overall rotation of the adduct species. The contribution of transitional modes to TS partition function is evaluated via Monte Carlo integration of classical phase space. The rate constant by energy- and total-angular-momentum-resolved microcanonical VRC-VTST (which we will abbreviate in this article as E, J, μ VT) for an association reaction is then given as

$$k_{\text{outer}} = g_e \frac{Q_{\text{trans}}^{E, J, \mu \text{VT}}}{h Q_{\text{rot}}^{R1} Q_{\text{rot}}^{R2} \Phi_{\text{rel}}}, \quad [4]$$

where g_e is the ratio of the electronic partition function of the TS to that of the reactants; Q_{rot}^{R1} and Q_{rot}^{R2} are the rotational partition functions of the reactants; Φ_{rel} is the relative translational partition function per unit volume of the separated reactants; and $Q_{\text{trans}}^{E, J, \mu \text{VT}}$ is the partition function for the translational modes, as given by

$$Q_{\text{trans}}^{E, J, \mu \text{VT}} = \left(\frac{1}{\sigma^\ddagger} \right) \iint_{E, J} N_{\text{trans}}^{E, J, \mu \text{VT}}(E, J) e^{-E/k_B T} dE dJ, \quad [5]$$

where $\sigma^\ddagger$ is the rotational symmetry number of the TS (see further discussion in *SI Appendix* for its definition); and $N_{\text{trans}}^{E, J, \mu \text{VT}}(E, J)$ is the number of vibrational-rotational accessible states of the generalized TS calculated with the optimum s for each energy E and total angular momentum J . Details of the calculation of $N_{\text{trans}}^{E, J, \mu \text{VT}}(E, J)$ and the integral have been given previously (14, 58).

Canonical unified statistical theory. The canonical unified statistical (14, 27) (CUS) theory provides a way to include the effect of two TSs in series. [For TSs in parallel, one would use MS-VTST or MP-VTST (14, 59).] The CUS rate constant k_{CUS} can be expressed in terms of k_{outer} (the rate constant for passing through the outer TS), k_{inner} (the rate constant for passing through the inner TS), and k_{complex} (the rate constant for maximum flux through the intermediate between the two TSs) as follows (14):

$$\frac{1}{k_{\text{CUS}}} = \begin{cases} \frac{1}{k_{\text{outer}}} + \frac{1}{k_{\text{inner}}} - \frac{1}{k_{\text{complex}}}, & \Delta G_T^{\text{VTS, inner, o}} > 0 \\ \frac{1}{k_{\text{outer}}}, & \text{otherwise} \end{cases} \quad [6]$$

We obtained k_{outer} and k_{inner} by VRC-VTST and RP-VTST, respectively, and k_{complex} is approximated using Eq. 3 by assuming the local stable complex between the two TSs is a generalized TS and therefore omitting its 85-cm⁻¹ vibration that has a normal mode character similar to the imaginary vibration of the inner saddle point (*SI Appendix*, Fig. S3). We have also replaced the DFM potential energies with the W3X-L benchmark ones in the calculation of k_{complex} .

Equilibrium Constants. We also calculate the K_c so that the reverse rate constant k_{-1} can be obtained from the forward rate constant k_{CUS} by detailed balance as

$$k_{-1} = k_{\text{CUS}}/K_c. \quad [7]$$

The equilibrium constant for the association reaction was computed by

$$K_c(T) = \left(\frac{k_B T}{P^0} \right) \exp \left(-\frac{\Delta G_T^0}{k_B T} \right), \quad [8]$$

where ΔG_T^0 is the standard-state free energy of reaction at temperature T . In this calculation, the DFM potential energies

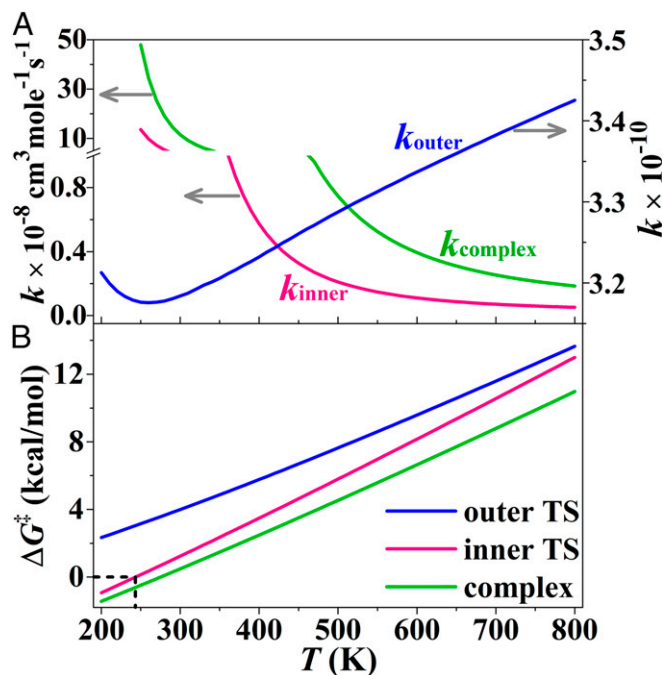


Fig. 3. (A) Rate constants k_{outer} and k_{inner} , reactive flux coefficients k_{complex} and (B) free energies of activation for the outer TS, the inner TS, and the complex. The values for the outer TS are obtained from k_{outer} , and the potential energies for the inner TS and the complex are corrected with W3X-L benchmark results.

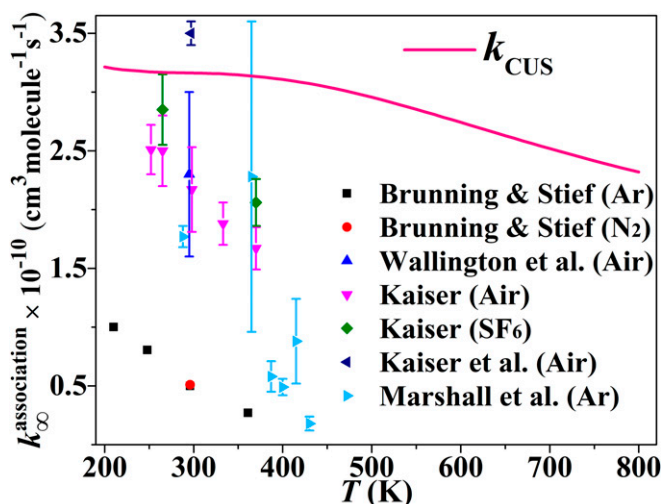


Fig. 4. The calculated and experimental high-pressure limit association rate constants for the reaction $\text{Cl} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_2\text{Cl}$. The bath gas used in each experiment is given in parentheses.

are replaced with the W3X-L benchmark results for better accuracy.

One complication in calculating $\Delta G_0^\ddagger$ is the existence of two conformations of the product β -chlorovinyl, as shown in Fig. 1, in particular *trans* β -chlorovinyl (P1a) and *cis* β -chlorovinyl (P1b). The isomerization between two conformations via $\text{C}\equiv\text{C}-\text{H}$ bending has a modest barrier (3.94 kcal/mol by W3X-L, 3.90 by WMS, 3.80 by M06-2X/may-cc-pVTZ; 2.68 kcal/mol if we add zero-point energy to get an enthalpy of activation at 0 K) so we treat the two conformations as being in equilibrium. Two procedures were used to treat the conformations for the calculation of equilibrium constants.

The first procedure is to treat the conformations independently. The total equilibrium constant for reaction 1 is then the sum (60, 61):

$$K_c = K_{c,\text{trans}} + K_{c,\text{cis}}. \quad [9]$$

The second procedure recognizes that the $\text{C}\equiv\text{C}-\text{H}$ bending mode of β -chlorovinyl is a double-well potential and treats the two conformations as a single isomer with a double-well vibration. The partition function for this mode is obtained by solving for the anharmonic energy levels of the double-well potential by using the WKB method.

Data Availability Statement. All data discussed in the paper are available in the main text or *SI Appendix*.

Results and Discussion

Potential Energy Surface. The potential energy surface (PES) for reaction 1 is shown in Fig. 1. A more complete diagram including some higher-energy species is given in *SI Appendix*, Fig. S4. The stationary points are consistent with previous theoretically studies (36, 37). To validate the DFM, we first optimized each stationary geometry using a given DFM; then we did single-point calculation with W3X-L for each DFM-optimized geometry. In *SI Appendix*, Tables S1 and S2, we give the comparisons for the M06-2X and M08-HX functionals with three triple- ζ basis sets. The comparison shows that M06-2X/may-cc-pVTZ DFM gives the smallest MUE (0.66 kcal/mol) for the relative energies of the stationary points and also gives a small mean signed error (−0.47 kcal/mol).

We have previously shown that M06-2X and M08-HX yield reasonably accurate geometries for TSs (62). *SI Appendix*, Tables S1 and S2 show that the high-level calculations give almost the

same energy for structures optimized with different DFMs; this adds confidence that any errors in the density-functional geometries do not cause a large error in the present results.

To further validate the DFM, we performed a relaxed scan along the association path of the reaction $\text{Cl} + \text{C}_2\text{H}_2$ [doing constrained optimization at each fixed $\text{C}(1)-\text{Cl}$ distance] with M06-2X/may-cc-pVTZ. Benchmark energies were calculated by W3X-L and WMS for each geometry along the relax scan curve. The DFM curve and the benchmark curves are shown in Fig. 2 with the zero of energies at the separated reactants. The energy deviations of the selected DFM compared with W3X-L are less than 1.5 kcal/mol in the whole range of 2.5–5.0 Å, and the deviations in the important VST region (see the contour map given in *SI Appendix*, Fig. S5) of 4.0–4.5 Å (where the loose outer TS is located) are even smaller (less than 0.6 kcal/mol). On the bases of the tests described in this paragraph and the previous two paragraphs and of the affordable cost of M06-2X/may-cc-pVTZ, we decided that M06-2X/may-cc-pVTZ provides a good compromise between accuracy and computational cost for our direct dynamics calculations.

The difference between the two high-level composite methods is less than 0.4 kcal/mol for the geometries of the relaxed scan, which lends further support to the accuracy of both benchmark calculations and—which can be very important for future work—shows that the WMS methods can give quite accurate results at a relatively low computational cost. The average ratio of the computer time for WMS as compared to W3X-L is 0.076, which is averaged over 34 calculations of the relaxed scan (Fig. 2).

In *SI Appendix*, Table S3, we list the unscaled harmonic vibrational frequencies of the stationary points using M06-2X/may-cc-pVTZ. In the calculations, these were scaled to obtain more accurate zero-point energies.

Table 1. Calculated (k_{CUS}) and experimental high-pressure limit association rate constants $k_{\infty}^{\text{association}}$ (in the units of $10^{-10} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$) at various temperatures

Ref.	Authors, year	T, K	Experiment	k_{CUS}	$k_{\text{CUS}}/\text{Exp.}$
30	Brunner and Stief (1985)	210	1.00*	3.20	3.2
		248	0.80*	3.17	4.0
		296	0.50*	3.16	6.3
		361	0.27*	3.14	12
32	Wallington et al. (1990)	295	$2.3 \pm 0.7^\dagger$	3.16	1.4
33	Kaiser (1992)	252	$2.5 \pm 0.2^\dagger$	3.17	1.3
		265	$2.6 \pm 0.5^\dagger$	3.17	1.2
		298	$2.2 \pm 0.4^\dagger$	3.16	1.4
		333	$1.9 \pm 0.2^\dagger$	3.15	1.7
		370	$1.9 \pm 0.4^\dagger$	3.13	1.6
34	Kaiser and Wallington (1996)	297	$3.5^\S$	3.16	0.9
63	NASA (2003)	250	$2.5 \pm 0.4^\P$	3.17	1.3
		298	$2.1 \pm 0.4^\P$	3.16	1.5
		370	$1.7 \pm 0.4^\P$	3.13	1.8
		288	$1.77 \pm 0.18^\dagger$	3.16	1.8
36	Gao et al. (2007)	365	$2.28 \pm 2.64^\dagger$	3.14	1.4
		387	$0.58 \pm 0.26^\dagger$	3.12	5.4
		400	$0.49 \pm 0.14^\dagger$	3.11	6.3
		415	$0.88 \pm 0.36^\dagger$	3.09	3.5
		430	$0.18 \pm 0.06^\dagger$	3.07	17
64	NASA (2015)	250	$2.5 \pm 0.4^\P$	3.17	1.3
		298	$2.2 \pm 0.4^\P$	3.16	1.4
		370	$1.9 \pm 0.4^\P$	3.13	1.6

*High-pressure limit.

[†]High-pressure limit with 2σ error bars.

[‡]Range includes high-pressure limits obtained with two different bath gases.

[§]High-pressure limit obtained with best fit to data. Another fit gave 2.0.

[¶]NASA recommendation.

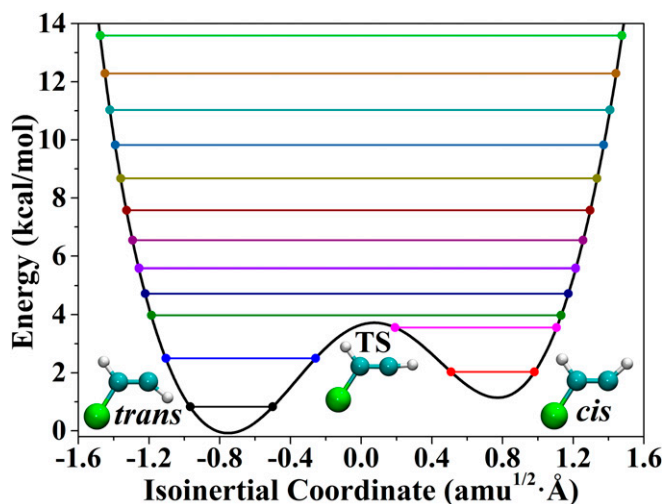


Fig. 5. The double-well potential for the C≡C-H bending in β-chlorovinyl calculated by M06-2X/may-cc-pVTZ and the associated anharmonic vibrational energy levels obtained by WKB method. Zero of the x axis is linear C≡C-H with negative and positive coordinates correspond to *trans* and *cis* conformation, respectively.

Association Rate Constants. The rate constants k_{outer} , k_{inner} , and k_{complex} (the data are given in SI Appendix, Table S4) are plotted in Fig. 3A, and the corresponding free energies of activation for the three rate constants are given in Fig. 3B. (Tabular presentations of rate constants are available in SI Appendix, Tables S4 and S5.) The temperature dependence for k_{outer} is negative for $T < \sim 260$ K and positive at higher T , while the T dependences of k_{inner} and k_{complex} are negative at all T . In SI Appendix, Table S4, we show that for $T < 450$ K, the outer TS yields a rate smaller than the inner one by a factor of 10 (450 K) to 3,700 (200 K); however the inner TS reduces the rate constant by 19, 18, 25, and 32% at 500, 600, 700, and 800 K, respectively.

The experimental measurements do not go to high enough pressure to directly measure the high-pressure limit, but three groups have attempted to extract the high-pressure limit by fitting the pressure dependence with theoretical guidance. The inferred high-pressure rate constants are given in Table 1 (30, 32–34), and some of them are plotted in Fig. 4; the table and figure include comparisons to our CUS rate constants. Table 1 also includes the 2003 and 2015 NASA recommendations (63, 64), which are based on the measurements of Kaiser (33) and Kaiser and Wallington (34). All of the experimental evaluations indicate the high-pressure limit rate constant has a negative temperature dependence in the available experimental temperature range of 210 to 430 K, and this is also found in the present calculations. Brunning and Stief (30) and Gao et al. (36) predicted a significant decrease in the higher-temperature part of the range, but a much smaller decrease is seen in the work of Kaiser (33) or the present calculations. Our calculated k_{CUS} rate constants show only an $\sim 3\%$ decrease over the experimental temperature range.

We fit the calculated rate constants to our previously (59, 65) recommended four-parameter function, which gives

$$k_{\text{CUS}} = 2.966 \times 10^{-9} \text{ cm}^3 \text{ molecule}^{-1} \cdot \text{s}^{-1} \left(\frac{T + 225.34}{300} \right)^{-1.654} \exp \left(\frac{0.698(T + 225.34)}{R(T^2 + 225.34^2)} \right),$$

[10]

where $R = 1.9872 \times 10^{-3} \text{ kcal mol}^{-1} \cdot \text{K}^{-1}$.

Association Equilibrium Constant. The bending potential that connects the two β-chlorovinyl conformations is a double-well potential. This anharmonic property increases the $\text{C}_2\text{H}_2\text{Cl}$ density of states (37) and therefore increases the association equilibrium constants. To study the influence of this anharmonic vibration, we used the WKB approximation to solve the one-dimensional Schrödinger equation in isoinertial (54, 66) coordinates for the vibrational energy levels of the C≡C-H bending mode; the *trans* β-chlorovinyl was used, and the double-well bending potential calculated by a relaxed scan with M06-2X/may-cc-pVTZ and fitted to eighth-order polynomial, as shown in Fig. 5, which also shows the first few energy levels (the energy levels and inner and outer turning points for each level are listed in SI Appendix, Table S6). The distance in isoinertial coordinates between two geometries along the relaxed scan is obtained by orienting the points along the path using Chen's method (67).

There are three vibrations (350, 694, and 821 cm^{-1} , SI Appendix, Fig. S6) of the *trans* β-chlorovinyl that show C≡C-H bending character. (All vibrational frequencies mentioned in this paragraph are the unscaled ones.) The normal modes and harmonic frequencies (SI Appendix, Figs. S6–S8) of these vibrations indicate that the 694- cm^{-1} vibration is the one corresponding to the C≡C-H double-well potential. This compares well to the harmonic frequency of 677 cm^{-1} on our fitted bend. We may compare this aspect of our calculations to the work of Gao et al. (36), who also computed the anharmonic vibrational levels of the C≡C-H bending for thermochemistry calculations; the C≡C-H bending frequency of the *trans* β-chlorovinyl in their study was 364 cm^{-1} as compared to 694 cm^{-1} here. Consequently we obtained quite different energy levels than Gao et al.

To compute partition functions and free energies of the product, we calculated the partition function for the 694- cm^{-1} mode by summing Boltzmann factors of the WKB energy levels, and we treated the other vibrations (scaled by 0.971 as discussed in SI Appendix) by the quasiharmonic oscillator approximation.

The computed equilibrium constants at various temperatures by the above-mentioned two procedures are listed in Table 2 (see SI Appendix, Table S7, for results at more temperatures) and compared with available experimental results from Gao et al. (36). The results with the anharmonic bend are larger, which means the anharmonic treatment for the C≡C-H bending has indeed increased the density of states. Although we have corrected the DFM potential energies with W3X-L benchmark values and included the anharmonicity of the C≡C-H anharmonic vibration, our computed values are higher than the experimental results by factors of 8–34. Further experimental and theoretical studies would be desirable to resolve the discrepancy, as there are no other available results for comparing.

Summary

The barrierless addition of Cl atoms to the C_2H_2 molecule has two VTSSs in series along the reaction path. We treated one by

Table 2. Calculated and experimental association equilibrium constants K_c (in the units of $\text{cm}^3 \text{ molecule}^{-1}$)

T , K	K_{exp}^*	$K_{c,1}^\dagger$	$K_{c,1}/K_{\text{exp}}$	$K_{c,2}^\ddagger$	$K_{c,2}/K_{\text{exp}}$
365	1.59×10^{-14}	3.33×10^{-13}	20.94	4.17×10^{-13}	26.24
387	6.17×10^{-15}	7.62×10^{-14}	12.35	9.51×10^{-14}	15.41
400	4.41×10^{-15}	3.44×10^{-14}	7.81	4.29×10^{-14}	9.72
415	2.69×10^{-15}	1.46×10^{-14}	5.44	1.82×10^{-14}	6.76
430	1.34×10^{-15}	6.62×10^{-15}	4.94	8.20×10^{-15}	6.12

*Experimental values from ref. 36.

†Calculated by assuming the two β-chlorovinyl conformations are independent.

‡Calculated using the WKB anharmonic energy levels for the C≡C-H bending mode.

VRC-VTST and the other by RP-VTST. The CUS model is used to account for the statistical branching probability for the complex to redissociate to reactants. The CUS rate constants successfully agree with the negative temperature dependence of the experimental high-pressure limit association rate constants and are in reasonable agreement with some of the experimental results; they cannot agree with all of the experiments since the experiments differ among themselves, but all of the experimental results indicate a steeper temperature dependence than theory. The association equilibrium constants have also been predicted based on the calculated free energy of reaction, including a treatment of the anharmonicity of the double-minimum $\text{C}\equiv\text{C}-\text{H}$

bending mode. This anharmonic treatment increases the calculated equilibrium constant, which is larger than the experimental ones. Possible reasons for the remaining differences of experimental and theoretical rate constants include the simplified treatment of reactant and product anharmonicity, the not easily estimated uncertainties in the experiments, and the sensitivity of the results to the details of the electronic structure calculations.

ACKNOWLEDGMENTS. This work was supported in part by the US Department of Energy, Office of Basic Energy Sciences, under Grant DE-SC0015997, and by the National Natural Science Foundation of China under Grant 51536002. L.Z. was supported by a scholarship from the China Scholarship Council (201706120185).

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